AMD DEVELOPER HACKATHON · ACT III · TRACK 2

SafeSignal

Helping people and pharmacy staff report a suspected medicine side effect completely, with a person reviewing every field.

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Demo only · fictional data · not medical advice

The problem

- Side-effect reports are often incomplete: medicine, dose, dates or outcome are missing.
- The reporter then has to be asked again, which slows everything down.
- Free text is easy to write but hard to turn into a valid structured report.
- Urgent wording can be buried in a long description.

The solution

1. Describe
The person writes what happened in their own words.

2. Extract
An open model pulls out the facts and quotes the words behind each field.

3. Check
Plain rules flag what is missing and how urgent it looks, with a rule id for each reason.

4. Review
The person edits fields, confirms the review, and downloads a structured draft.

Nothing is sent anywhere. There is no send button, on purpose.

How it works on AMD

- Model: Gemma (`google/gemma-3-27b-it` planned; 12b or an open model as a fallback, stated honestly).
- Serving: vLLM on ROCm, on one AMD Instinct MI300X in AMD Developer Cloud.
- One OpenAI-compatible endpoint; only environment variables change between backends.
- **The model only extracts.** Completeness and urgency come from deterministic rules, so an injected prompt cannot lower urgency.
- The page always shows what is really answering: live on AMD, a dev stand-in, or a labelled replay of a real AMD run.

Demo

- Pick a fictional sample; see extracted fields with the quotes they came from.
- See missing items and the questions to ask; edit any field.
- Red-flag sample: emergency wording with rule ids. The injection sample stays emergency.
- Review box, then download the structured report.
- Demo URL: [ADD APPLICATION URL] Video: [ADD VIDEO LINK]

Results

Values come from `npm run bench` on the real AMD endpoint. Nothing is filled in until measured.

Measure	Value
Hardware / software	[rocm-smi, vllm --version]
Model served	[model actually served]
Extraction quality on sample cases	[from bench]
Latency per report	[from bench]
Throughput at several concurrency levels	[from bench]

Limits and next steps

- Triage rules are keyword based and English only: a transparent aid, not a clinical tool.
- All data is invented; the demo must not receive real patient information.
- Next: more languages, review of the rules by pharmacovigilance professionals, field testing with pharmacy staff.
- Next: map the draft to the format a regulator expects, still with human review before any submission.

Open source, MIT licence · Repository: github.com/iancuileana83-lab/safesignal